

Computer Boom... Environmental Bust!

In an age that worships faster computing and sci-fi devices, Mother Nature goes ignored. Here's how technology bytes its way into pollution

When the surge in computing power became apparent, it was thought that an industry, based on efficiency, was born. In the last couple of decades India has seen the installed base of computers rise from zero to millions. With the rapid progress of technology, most of these computers become obsolete in 2-3 years, adding to the pile of garbage on the street corner. This is an environmental nightmare just waiting to happen. This is further aggravated by the fact that little was known about technology, and establishing lifecycles for any product is difficult.

Researchers in current times have dissected the issue, resulting in a reality check for most people. This has also highlighted several other issues that were previously ignored.

Microchip pollution

The production of computer components requires hundreds of materials that are potentially harmful to the environment. The basic building block of any unit in your PC from modem to motherboard is

the Printed Circuit Board (PCB). These boards are laminated boards on which various components are mounted. Circuit tracks are etched onto this board, which results in a hardwired circuit.

Silicon and Gallium Arsenide are the main materials used in the manufacture of these components. These components are then treated to adjust their electrical properties. Now, Silicon or Gallium Arsenide crystals are formed, which are then shaped and sliced to thin wafers. These are covered with multi-resistant chemicals. This step of the manufacturing process can pollute the surrounding environment. A process called doping is carried out using toxic gases, and an unsafe solvent is used to wash away the multi-resistant chemical.

The surface of the PCB must be kept clean using Chlorofluorocarbons (CFCs). The volume of chip production is extremely high. Together this means that chip production facilities use up a large amount of CFCs.

The effect of these processes on the environment is adverse—air and ground water pollution are the two obvious side effects. The use of CFC's depletes the ozone layer and the leakage of such solvents causes water contamination.

The amount of resources required to make a small memory chip is equally stunning. According to the research done by the United Nations University in Tokyo; the cost of a 2 gms memory chip with capacity of 32 MB requires 1,400 grams of materials—fossil fuels, chemicals and the raw materials. This is a ratio of about 700 times the material used for the chip, as compared to a ratio of 2 for cars. If 1,400 gms of materials does not seem much, think of the number of integrated chips used in a PC.

Major manufacturers, such as AMD and Intel, are implementing designs →